Annihilation 3

From the first section we have

$$\sum_{ab} \mathcal{M}_1{}^{\mu\nu} \mathcal{M}^*_{1\mu\nu} = \frac{e^4}{(t-m^2)^2} \left(8tu - 24m^2t - 8m^2u - 8m^4\right)$$

$$\sum_{ab} \mathcal{M}_1{}^{\mu\nu} \mathcal{M}^*_{2\nu\mu} = \frac{e^4}{(t-m^2)(u-m^2)} \left(8m^2s - 32m^4\right)$$

$$\sum_{ab} \mathcal{M}_2{}^{\nu\mu} \mathcal{M}^*_{2\nu\mu} = \frac{e^4}{(u-m^2)^2} \left(8tu - 8m^2t - 24m^2u - 8m^4\right)$$

We also have by computer algebra

$$\sum_{ab} \mathcal{M}_1{}^{\mu\nu} \mathcal{M}^*_{1\mu\nu} = \frac{e^4}{(t-m^2)^2} \left(32(p_1 \cdot p_3)(p_1 \cdot p_4) + 32m^2(p_1 \cdot p_3) - 32m^4\right)$$

$$\sum_{ab} \mathcal{M}_1{}^{\mu\nu} \mathcal{M}^*_{2\nu\mu} = \frac{e^4}{(t-m^2)(u-m^2)} \left(16m^2(p_1 \cdot p_2) - 16m^4\right)$$

$$\sum_{ab} \mathcal{M}_2{}^{\nu\mu} \mathcal{M}^*_{2\nu\mu} = \frac{e^4}{(u-m^2)^2} \left(32(p_1 \cdot p_3)(p_1 \cdot p_4) + 32m^2(p_1 \cdot p_4) - 32m^4\right)$$

Eigenmath script